

COS30045

LAB 4.1 Design Studio



Overview

In this lab you will be given a sample data set and asked to identify the different data and attribute types. You will also think about some questions about this data set that might be answered by a visualisation.

[ardd_fatalities_Jan2020_0.xlsx](#) (download from Canvas)

Download and review this data set before attempting this exercise.

1 Interpreting the data set

Complete the LAB 4.1 Quiz.

2 Visualisation Design

Think of three questions you would like to answer with that require a data visualisation.

For each data question you will need to consider the following:

Which data attributes (columns) do you need to answer this question?

Do you need to transform any of the data?

Does the data type change when you transform the data? If so how.

Make a sketch of how you think your visualisation might look and add to this document.

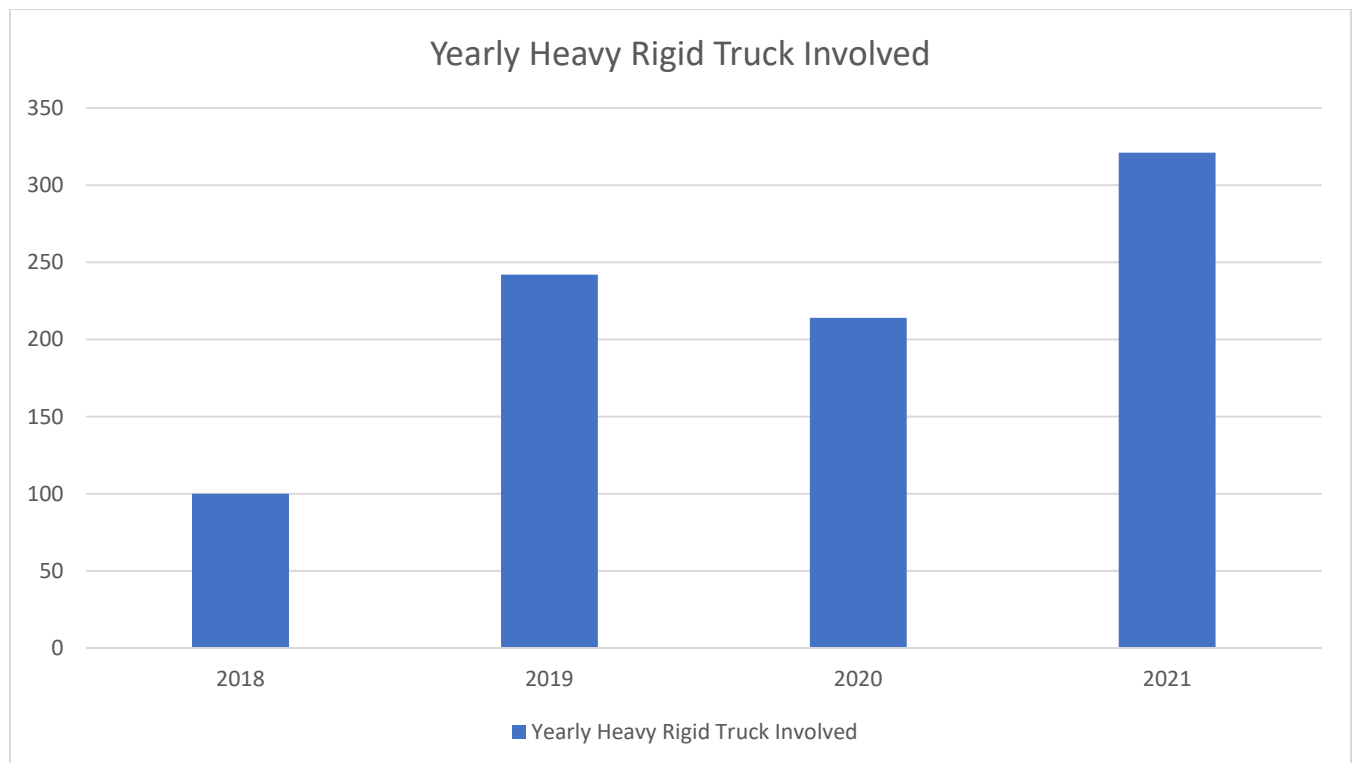
Your Question 1

Question: What are the Yearly number of road fatalities that has heavy rigid truck involved?

Column Needed: Year, Heavy Rigid Truck Involved

Transform Data: Yes, only take the row with value Yes in column Heavy Rigid Truck Involved and group by Year.

Data Type Change: There will be quantity/crash count which count by yearly have how many, which will be integer value.



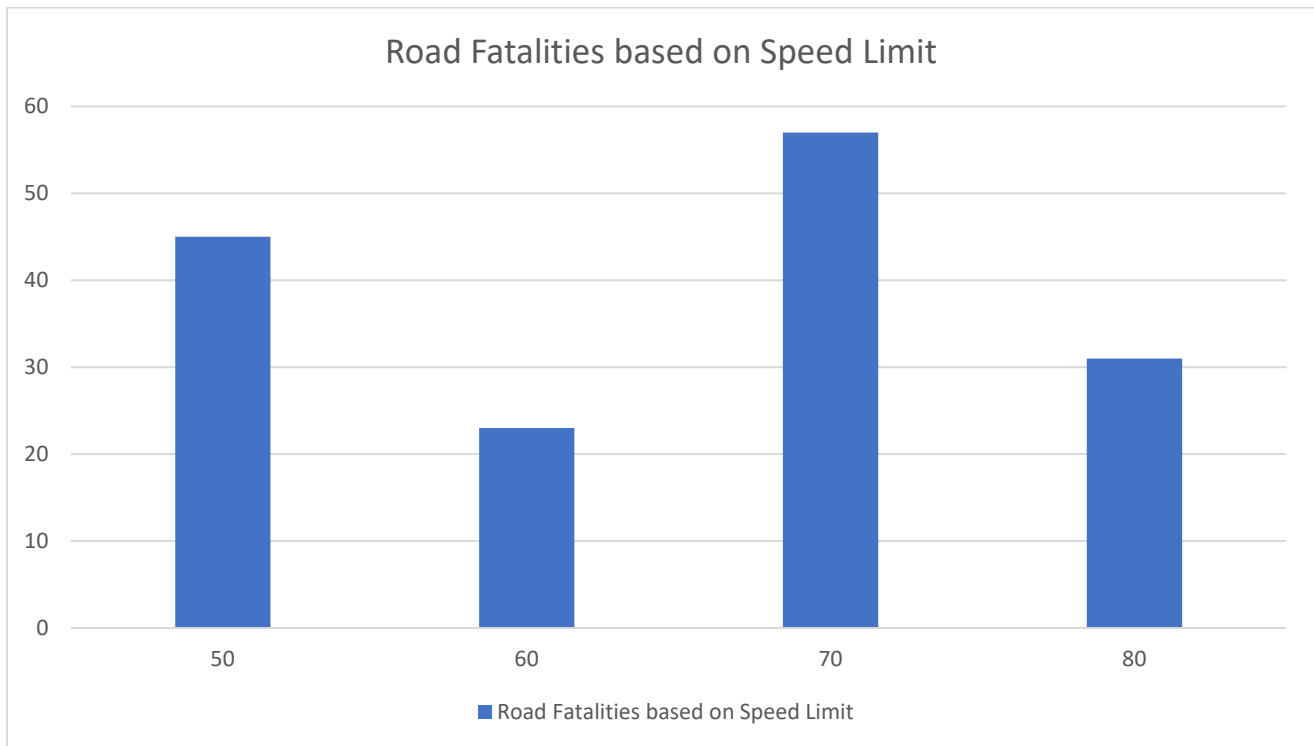
Your Question 2

Question: What are the number of road fatalities for different Speed Limit?

Column Needed: Speed Limit

Transform Data: No

Data Type Change: Yes, need to group the speed limit and count how many same speed limit are there, with integer data type.



Your Question 3

Question: What time of day has the highest number of road fatalities?

Column Needed: Time

Transform Data: Yes, Convert the Time column into time intervals and count crashes in each period.

Data Type Change: Yes, Transform Time from specific time points into categorical intervals.

